mathematical Formulation from Scratch

$\hat{y} = mx + c$ — For simple linear regression

$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \ldots + \beta_n x_n$ — For multiple linear regression

Suppose,
we have a dataset having columns as CGPA, IQ, Gender and target Package
so $\hat{y}$ becomes

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

So to calculate predicted y ($\hat{y}$) with the help of x_1, x_2 & x_3

$$\begin{bmatrix} \hat{y}_1 \end{bmatrix} \quad \begin{bmatrix} \beta_0 & \beta_1 x_{11} & \beta_2 x_{12} & \beta_3 x_{13} \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} \hat{y}_2 \\ \vdots \\ \hat{y}_{100} \end{bmatrix} = \begin{bmatrix} \beta_0 & \beta_1 x_{21} & \beta_2 x_{22} & \beta_3 x_{23} \\ \vdots & \vdots & \vdots & \vdots \\ \beta_0 & \beta_1 x_{1001} & \beta_2 x_{1002} & \beta_3 x_{1003} \end{bmatrix}$$

	1	2	3
x_1	CGPA	IQ	Gender
x_2			
$\vdots$			
x_{100}			

n Rows , m columns

$$\hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} \beta_0 & \beta_1 x_{11} & \beta_2 x_{12} & \cdots & \beta_m x_{1m} \\ \beta_0 & \beta_1 x_{21} & \beta_2 x_{22} & \cdots & \beta_m x_{2m} \\ \vdots & & & & \\ \beta_0 & \beta_1 x_{n1} & \beta_2 x_{n2} & \cdots & \beta_m x_{nm} \end{bmatrix}$$

$$= \overset{A}{\begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1m} \\ 1 & x_{21} & x_{22} & \cdots & x_{2m} \end{bmatrix}} \overset{B}{\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}}$$

$$\begin{bmatrix} \vdots & & & \\ 1 & x_{n1} & x_{n2} \cdots & x_{nm} \end{bmatrix} \begin{bmatrix} \vdots \\ \beta_m \end{bmatrix}$$

$\Rightarrow$ which is equal to matrix multiplication.

$$\Rightarrow \hat{y} = X\beta$$

$$e = y - \hat{y} \Rightarrow \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} - \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}$$

$$E = e^T e$$

$$= \begin{bmatrix} (y_1 - \hat{y}_1) & (y_2 - \hat{y}_2) \cdots & (y_n - \hat{y}_n) \end{bmatrix} \begin{bmatrix} (y_1 - \hat{y}_1) \\ (y_2 - \hat{y}_2) \end{bmatrix}$$

$$\left[(y_n - \hat{y}_n) \right]$$

$$= (y_1 - \hat{y}_1)^2 + (y_2 + \hat{y}_2)^2 + \ldots + (y_n - \hat{y}_n)^2$$

$$E = \sum (y_i - \hat{y}_i)^2$$

$$E = e^T e = (y - \hat{y})^T (y - \hat{y})$$

$$= (y^T - \hat{y}^T)(y - \hat{y})$$

$$= \left[y^T - (X\beta)^T \right] (y - X\beta)$$

$$= y^T y - y^T X\beta - (X\beta)^T y + (X\beta)^T X\beta$$

$$= y^T y - 2y^T X\beta + \beta^T X^T X\beta$$

$$\frac{dE}{} = \frac{d}{} \left[y^T y - 2y^T X\beta + \beta^T X^T X\beta \right] = 0$$

$$\frac{d}{d\beta} \quad \frac{d}{d\beta}$$

$$= 0 - 2y^T x + \frac{d}{d\beta}(\beta^T x^T x \beta) = 0$$

$$= -2y^T x + 2x^T x \beta^T = 0$$

$$\Rightarrow \cancel{2} x^T x \beta^T = \cancel{2} y^T x$$

$$\beta^T = y^T x (x^T x)^{-1}$$

$$(\beta^T)^T = [y^T x (x^T x)^{-1}]^T$$

$$\beta = [(x^T x)^{-1}]^T (y^T x)^T$$

$$= [(x^T x)^{-1}]^T x^T y$$

$$= (x^T x)^{-1} x^T y$$

$$\boxed{\beta = (x^T x)^{-1} x^T y}$$